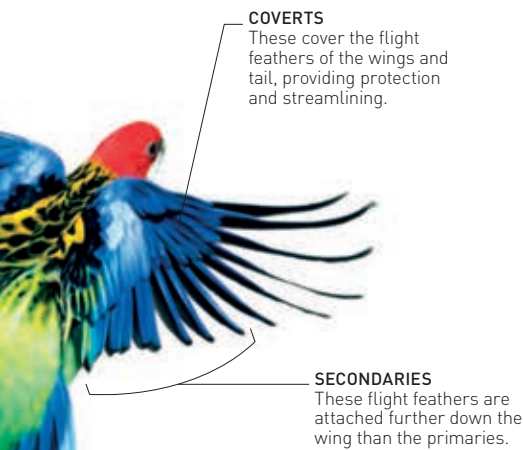
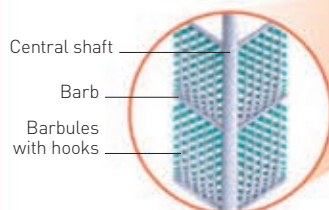




FEATHER STRUCTURE

Contour feathers have stiff shafts in the centre, with vanes on either side. The vanes are made up of thin branches called barbs. These have smaller branches of barbules with hooks that hold it all together.



CLOSE-UP

This complex but strong structure holds the feather together.



HOW BIRDS FLY

A bird uses strong breast muscles to flap its wings and fly. As it flaps them, it increases lift, which moves the bird forwards and upwards.



STAYING UP

Once in the air, the bird's wings allow it to glide, soar, change direction, and slow down to land.

DUCKS, GEESE, AND SWANS

These birds spend a lot of time in water, so their feathers are covered with an oily substance that keeps them waterproof.



SHOREBIRDS, SEABIRDS, AND WADING BIRDS

These birds have feathers that help with waterproofing and warmth. They tend to have dark feathers on their backs and white ones on their chests, to provide camouflage.



THE BIRD WITH THE MOST FEATHERS IS THE WHISTLING SWAN: IT HAS UP TO 25,000 FEATHERS IN WINTER

PERCHING BIRDS

The stiff tail feathers of many of these birds help them to balance while perching on small branches and garden fences.



OTHER BIRDS

There are many families of birds. Each family has feathers suited to its habits and lifestyle. Doves and pigeons create a white powder called feather dust to keep their feathers waterproof, unlike kingfishers, which use preening oil for waterproofing their feathers. Ostriches don't fly, so don't have any contour feathers, only soft down feathers.

